Isolation of single nuclei from postmortem fresh frozen human brain tissue and immunostaining for NeuN

Cobos Lab, October 2019

(based on Krishnaswami et al., Nat Protoc. 2016, 3:499-524)

Buffer preparation (All solutions should be RNase-free for single-soma RNAseq experiments):

Isolation Medium #1 (IM1), 45 ml

Prepare in a 50 ml Falcon tube and store at 4ºC up to 6 months.

7500 µl	1.5M Sucrose	(final concentration 250mM)
1125 µl	1M KCl	(25mM)
225 µl	1M $MgCl_2$	(5mM)
450 µl	1M Tris pH8	(10mM)
35.7 ml	RNase-free H_2O	

Homogenization Buffer (3 ml per sample).

Prepare FRESH and keep ICED or at 4ºC. Discard after use.

2925 µl	IM1	
3 µl	DTT 1mM	(final concentration 1µM)
30 µl	50x Protease Inhibitor	(0.5x)
15 µl	RNaseIN 40U/µl	(0.2U/µl)
30 µl	TritonX100 10%v/v	(0.1%)

Iodixanol dilutions

Prepare in a 50 ml Falcon tubes and store at 4ºC up to 6 months. Accuracy with Iodixanol and sucrose concentrations is critical.

Iodixanol medium (IDM), 45 ml

7500 µl	1.5M Sucrose	(final concentration 250mM)
6750 µl	1M KCl	(150mM)
1350 µl	1M $MgCl_2$	(30mM)
2700 µl	1M Tris pH8	(60mM)
26.7 ml	Nuclease-free H_2O	

Iodixanol 50%(v/v), 20 ml

16.7 ml Iodixanol 60%(v/v) + 3.3 ml IDM

Iodixanol 29%(v/v), 30 ml

14.5 ml Iodixanol 60%(v/v) + 15.5 ml IDM

Freezing Storage Buffer (FSB) 15ml

Prepare in a 50 ml Falcon tube and store at 4ºC up to 6 months.

1.665 ml	1.5M Sucrose	(final concentration 166.5mM)
75 µl	1M $MgCl_2$	(5mM)
150 µl	1M Tris pH8	(10mM)
13.1 ml	Nuclease-free H_2O	

Buffer for Immunostaining, 10ml

9400 µl	RNase-free PBS pH7.4	
500 µl	RNase-free 10% BSA	(final concentration 0.5%)
50 µl	1M $MgCl_2$	(5mM)
10 µl	DNAse I (2000U/ml)	(2U/ml)

Buffer for Antibody Incubation

Add to 1ml of Buffer for Immunostaining:

5 µl	RNaseIN 40U/µl	(0.2U/µl)

Collection medium for FACS (1ml; 0.2 ml/vial).

950 µl	RNase-free PBS pH7.4	
25ul	RNAse IN	
*	BSA 10% *After collecting!	(final concentration 1%)

Reagents

UltraPure™ distilled water, DNAse/RNAse-free (Thermo Fisher, 10977023)
Sucrose (Thermo Fisher, S25590B)
KCl (Sigma, 60142)
$MgCl_2$ (Sigma, M2670)
UltraPure™ 1M Tris-HCI Buffer, pH 8.0 (Thermo Fisher, T2944-1L)
DTT (Sigma, D9779-1G)
50x Protease Inhibitor (Roche, 11697498001)
RNaseIn (Lucigen, 30281-2)
DNAse I (NEB, M0303S)
BSA (Sigma, A7906)
Triton X-100 (ACROS Organics, 215680010)
Iodixanol 60% (v/v) (OptiPrep™ Density Gradient Medium; Sigma, D1556)

Protocol

- Material and tools needed: Forceps, spatula, blades, dounce tissue, cell strainers, petri dishes
- All solutions and materials should be RNase free and kept iced or at 4 ºC at all times
- Glassware and metal tools are sealed with aluminum foil and baked at 220 ºC for 6 h

1. Prepare Homogenization Buffer and cool it on ice.
   Pre-cool the dounce tissue grinder on ice
   (**Kimble Kontes all glass tissue grinder**, 7 mL tubes, pestle B, 0.02–0.056 mm clearance space between pestle and tube (Kimble, 885300-0007))
   Add 2.4 ml of Homogenization buffer to the dounce tissue grinder.

2. Collect the brain chunk (~100 mg) and transfer to a Petri dish on ice.
   Cut out into small pieces using a chilled scalpel or blade.

3. Transfer all pieces into the dounce tissue grinder. Homogenize the tissue with pestle B, on ice (**~30 strokes**). Check on hematocytometer while homogenizing and adjust the number of strokes (too few strokes – you will see cell clumps; too many strokes – you will get damaged nuclei).

4. Filter homogenate using **40 µm** Corning cell strainer to remove clumps.
   →Take sample for hemocytometer.

5. Transfer the homogenate into two precooled 1.5 ml Eppendorf tubes.
   Centrifuge at **1000g for 8 min** at 4ºC.

6. Slowly aspirate the supernatant from the side of the tube. Avoid disturbing the pellets, up to 50 µl of supernatant can be left.
   Gently **resuspend** each pellet in 225 µl (final volume) of cold Homogenization buffer and pool both tubes (final volume 450 µl).

7. Add an **equal** volume of cold **50%v/v iodixanol**. (Critical! Be very exact with volumes. 450 µl suspension + 450 µl 50% iod. Final iodixanol concentration is 25%) Gently pipette mix.

8. Add an equal volume of **29% iodixanol** (900 µl) into a 2 ml Eppendorf tube. Then, slowly layer off the 25% iodixanol suspension mix over the 25% iodixanol, without mixing them.
   Centrifuge at **13,500g for 20 min** at 4ºC.
   →Prepare buffer for immunostaining

9. After centrifugation, remove and discard the top myelin-rich debris layer (you can use a 1 ml pipette with the tip cut, or cotton swabs). Then, remove and discard the aqueous supernatant, without disrupting the **nuclei pellet**. Avoid contaminating with the top layer.
   Use a small amount **of buffer for Antibody Incubation** to resuspend the pellet and transfer the solution to a new tube.

10. Gently resuspend in 500 µl of **buffer for Antibody Incubation**. Incubate for 15 min, at 4ºC or iced, for **blocking** nonspecific staining.

    →Take sample for hemocytometer.

11. Take a sample for **unstained** control. Take sample for **2AB-only** (nonspecific binding control).

12. Add primary antibody (**Ms-a-NeuN, 1:1000**) and incubate on a rotator in cold room (4ºC) for 40 min.
    (NeuN antibody: Millipore #MAB377)

    → Note: use Eppendorfs coated with BSA for collection (to coat the Eppendorf tubes, fill them with 10% BSA solution in PBS for 5 min, rinse with PBS, and dry at 4°C overnight).

13. **Washing**. Add 700 µl of buffer for Immunostaining and invert several times. Centrifuge at 500g for 5 min at 4 ºC. Carefully transfer all the supernatant (don't leave any supernatant!It's ok to drag some pellet) to a new Eppendorf tube and resuspend the pellet in 400 µl of buffer for Antibody Incubation. Centrifuge the supernatant at 500g for 5 min at 4 ºC to recover non-pelleted nuclei, resuspend them in 200 µl of buffer for Antibody Incubation and pool for a final volume of 600 µl.

14. Take samples for single color staining.

15. Add secondary antibody (**G-a-Ms Alexa Fluor 647, 1:1000**) and Hoechst 33258 (5 µg/ml) to the same tube. (**Hoechst 1:1000 of 2,5 mg/ml stock**).

    →**Check** the quality of the sample on hemocytometer. For 100 mg of cerebral cortical tissue, should have ~ 1-2 Million nuclei in ~0.6 mL.

16. Transfer to 7ml culture tubes. Place on ice and bring them to the FACS facility for sorting.

17. Collect in BSA coated Eppendorf's containing collection medium (containing ~1/5 of the expected final volume after collection).

18. Collect in BSA-coated Eppendorfs containing collection medium (volume of collection medium should be ~1/5 of the expected final volume after collection).
    Add BSA after collection for final 1% BSA final concentration (tested for 10x Chromium v2 and v3). Different Single Cell assays tolerate different BSA concentrations, but lowering it may increase nuclei aggregation.

<u>FACS sorting controls:</u>
Non-stained
2AB only
Single stainings